

Simulation of Materials for Next-Generation Energy Devices

Quantum in Silico

Quantathon CR 2026 — Challenge 3

July 24, 2026

1 Introduction and objective

The simulation of magnetic materials and phenomena such as phase transitions remains scientifically relevant because of its broad applicability. The transverse-field Ising model (TFIM) is the canonical testbed for this class of problems: it combines a ferromagnetic interaction between neighboring spins with quantum fluctuations controlled by a transverse field and, in the thermodynamic limit, a quantum phase transition at $h/J = 1$ [1,2].

The goal of the challenge is to implement the time evolution of these models on the Quantinuum H2 family, detect the reorganization of observables around the critical point, and validate every circuit against a classical reference method. This work contributes: (i) sparse and symmetry-sector ED; (ii) exact and local-Trotter evolution; (iii) adiabatic preparation and quench protocols; (iv) real executions on Nexus (H2-1LE and H2-Emulator); (v) systematic studies of Δt and N ; (vi) VQE, ZNE, and error detection with the Iceberg code; and (vii) the full two-dimensional Fermi–Hubbard extension.

2 Models and observables

2.1 One-dimensional TFIM

The implemented Hamiltonian is

$$H = -J \sum_{i=0}^{N-1} Z_i Z_{i+1} - h \sum_{i=0}^{N-1} X_i. \quad (1)$$

The first term favors alignment of neighboring spins (ferromagnetic order along z); the second is a transverse field that does not commute with the first and induces fluctuations that destroy that order. Their competition produces the transition at $h/J = 1$. We use $J = 1$ as the energy unit and study $h/J \in \{0.5, 1, 2\}$. Periodic boundaries restore translational invariance and reduce finite-size effects, making a chain of 6–8 spins a more faithful surrogate for the infinite system.

The main observables are

$$M_z = \frac{1}{N} \sqrt{\left\langle \left(\sum_i Z_i \right)^2 \right\rangle}, \quad (2)$$

$$M_x = \frac{1}{N} \langle \sum_i X_i \rangle, \quad M_{zz} = \frac{1}{N} \langle \sum_i Z_i Z_{i+1} \rangle. \quad (3)$$

The notation M_z is important: the code does not report the signed longitudinal magnetization, which vanishes exactly in a definite-parity finite-size state because of the $\mathbb{Z}_2$ symmetry of (1), but its RMS magnitude, which

distinguishes the ordered and disordered phases. M_x , in contrast, is a signed average because the field $-h \sum_i X_i$ explicitly breaks the symmetry and polarizes the ground state toward $+x$.

2.2 Two-dimensional Fermi–Hubbard

The spinful Fermi–Hubbard Hamiltonian on a periodic $L_x \times L_y$ lattice is

$$H = -t \sum_{\langle ij \rangle, \sigma} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (4)$$

with $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$. The first term is hopping: the amplitude t with which an electron moves to a neighboring site, i.e. the band kinetic energy; it favors delocalized states. The second is the on-site Coulomb repulsion U : the energetic penalty for two electrons (necessarily of opposite spin by Pauli exclusion) occupying the same site; it favors localized states, one per site. Thus U/t is the only physical parameter. Sweeping it goes from the non-interacting metal ($U/t = 0$) to the Mott insulator with emergent antiferromagnetic order ($U/t \gg 1$). We always work at half filling, $\langle N \rangle = N_{\text{sites}}$ and $S_z = 0$, the part of the phase diagram most difficult and most relevant to cuprates.

The reported observables are the mean double occupancy

$$\langle D \rangle = \frac{1}{N_{\text{sites}}} \sum_i \langle n_{i\uparrow} n_{i\downarrow} \rangle, \quad (5)$$

the staggered (antiferromagnetic) magnetization

$$m_s = \frac{1}{N_{\text{sites}}} \sum_i (-1)^{x_i+y_i} \langle (n_{i\uparrow} - n_{i\downarrow})/2 \rangle, \quad (6)$$

and the site density $\langle n_i \rangle$. $\langle D \rangle$ directly measures how much double occupancy the ground state tolerates: it is 0.25 in the uncorrelated limit (each spin has independent probability 1/2 on each site) and decreases toward zero as U grows. m_s measures Néel order. All three are diagonal in the computational basis under Jordan–Wigner, $n = (1 - Z)/2$, so they can be measured with one circuit in the Z basis—unlike the TFIM, where M_x requires a second circuit with a basis rotation.

3 Methodology

3.1 Exact diagonalization

Exact diagonalization (ED) was used as the performance reference. For quantum systems, ED provides the exact

solution of the Hamiltonian and serves as a benchmark for approximate methods such as Trotterization [3]. Since the system under study is a chain of $N = 8$ spins, ED is computationally feasible and is the most accurate classical method available at this scale [21].

The Hamiltonian and observables are built as sparse CSR matrices using sparse Kronecker products to reduce memory load. The ground state is obtained with a sparse eigensolver; for dynamics, the full matrix is diagonalized once and propagated:

$$|\psi(t)\rangle = \sum_n e^{-iE_n t} \langle E_n | \psi(0) \rangle |E_n\rangle. \quad (7)$$

This is exact and free of discretization error, and therefore provides the pattern against which Trotter error is measured.

For Fermi–Hubbard, full-space ED is infeasible beyond 2×2 : a 3×4 lattice has 12 sites, 24 qubits, and dimension $2^{24} \simeq 1.7 \times 10^7$. Since (4) conserves $N_\uparrow$ and $N_\downarrow$ separately, the project implements a sector engine (‘fh_sector.py’) restricted to the half-filled $S_z = 0$ sector, of dimension D^2 with $D = \binom{N_{\text{sites}}}{N_{\text{sites}}/2}$. Because the up modes occupy JW indices $0 \dots N-1$ and down modes $N \dots 2N-1$, the sector factorizes:

$$H = T \otimes I + I \otimes T + D_{\text{int}}. \quad (8)$$

Storing the state as a matrix $\Psi[i_\uparrow, i_\downarrow]$ reduces the matrix-vector product to $T\Psi + \Psi T^\top + D_{\text{int}} \circ \Psi$. For 3×4 , this reduces 2^{24} to 853,776 dimensions (5.1% of the full space). Time evolution within the sector uses a Krylov–Lanczos exponential step. Half filling is exact by construction, with no need for the chemical-potential trick $\mu = U/2$.

3.2 Trotterization

The Lie–Trotter formula separates H into $H_{ZZ} + H_X$ for the TFIM, or hopping plus interaction for Hubbard. Each part is easy to exponentiate because its terms commute, and the parts are alternated in short steps:

$$U_1(\Delta t) = e^{-iH_A \Delta t} e^{-iH_B \Delta t} = e^{-iH \Delta t} + O(\Delta t^2). \quad (9)$$

Repeated $r = t/\Delta t$ times, the accumulated global error is $O(\Delta t) = O(r^{-1})$. The symmetric formula inserts a half-step on either side:

$$U_2(\Delta t) = e^{-iH_A \Delta t/2} e^{-iH_B \Delta t} e^{-iH_A \Delta t/2} = e^{-iH \Delta t} + O(\Delta t^3), \quad (10)$$

with global error $O(\Delta t^2)$ [3,4]. Second order was selected for its higher accuracy without an excessive circuit-evaluation cost.

When steps are chained, the final half rotation of one step and the initial half rotation of the next act on the same qubit with nothing between them, so they fuse exactly through $R_x(a)R_x(b) = R_x(a+b)$. The resulting circuit has $r+1$ layers of R_x instead of $2r$: it is unitary-equivalent, one order more accurate, and uses fewer gates than the naive first-order version. For the $N = 6$ adiabatic sweep with 100 steps, fusion reduced the gate count from 1812 to 1218 and depth from 402 to 303 ($\sim 33\%$), without changing any observable (unitary difference verified at $\sim 10^{-15}$). This is the same idea of reducing gate count

through circuit identities that Quantinuum applies to fermionic SWAP gates [14].

The concrete TFIM gates are $R_X(-2h\Delta t)$ and $R_{ZZ}(-2J\Delta t)$; pytket provides native ‘ZZPhase’, so no $CX-R_z-CX$ decomposition is required. The ZZ rotations are grouped by edge-coloring the chain graph: edges of the same color do not share qubits and can run in the same layer. For a periodic chain this gives 2–3 color classes independently of N , hence the constant depth (4–5) measured in Table 4. For Fermi–Hubbard, each mapped term $e^{-i\theta P}$ is implemented with a pytket ‘PauliExpBox’. Terms are ordered so that the diagonal, mutually commuting interaction block is adjacent at the center of the symmetric sweep; the two halves fuse exactly, giving the literature structure $U_{\text{int}}U_{\text{hop}}U_{\text{int}}$.

3.3 Jordan–Wigner mapping

Electrons are fermions: exchanging two of them introduces a minus sign, whereas qubits do not do so natively. Jordan–Wigner (JW) mapping [6] assigns one qubit per fermionic mode ($|1\rangle = \text{occupied}$, $n = (1 - Z)/2$) and carries antisymmetry in a string of Z operators:

$$a_p = \left(\prod_{q < p} Z_q \right) \frac{X_p + iY_p}{2}, \quad (11)$$

so a hop between modes $p < q$ is

$$c_p^\dagger c_q + \text{h.c.} = \frac{1}{2}(X_p X_q + Y_p Y_q) \prod_{p < k < q} Z_k. \quad (12)$$

The cost is that the string becomes long when modes are far apart, and a 2D lattice has no natural linear ordering. We mitigate this in two ways: sites are traversed in a snake ordering, keeping neighbor strings short; and all up modes precede all down modes, so an up-spin hop never crosses down-spin modes. On-site interaction is diagonal and needs no string.

3.4 Two protocols: adiabatic and quench

In adiabatic preparation, the system starts in $|+\rangle^{\otimes N}$, exactly the ground state as $h \rightarrow \infty$, and the field is slowly ramped from $h_{\text{init}} = 4$ to the target value while J grows from 0 to 1. The adiabatic theorem guarantees that, if the ramp is slow compared with the energy gap, the system remains in the instantaneous ground state. This prepares the ground state and therefore reveals the phase transition.

In the quench protocol, the system starts in $|0 \dots 0\rangle$ and evolves under a fixed Hamiltonian. This protocol does not prepare anything; it directly measures Trotter time error against ED, because any discrepancy is attributable to the circuit rather than to the physics. It is the protocol used to verify the challenge’s $< 5\%$ criterion.

3.5 VQE: the ansatz is half the algorithm

VQE [8] seeks the ground state by minimizing

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle \quad (13)$$

over a parameterized circuit, with the circuit on the quantum device and the optimizer classically. It is an alternative to adiabatic preparation that does not suffer diabatic transitions near the critical point because it follows no time trajectory. Its weakness is that result quality depends strongly on the ansatz.

We used the Hamiltonian variational ansatz (HVA) [9], which reproduces the structure of the Hamiltonian itself—layers of $(ZZ\text{Phase}, R_x)$ for the TFIM and a colored Trotter circuit with one free angle per layer instead of two. This usually gives better results than a generic hardware-efficient ansatz with the same number of parameters.

For Fermi–Hubbard, the HVA must preserve the number of particles, so the optimizer acts only on the physical sector. This imposes that X and Y terms with the same spin are grouped together, while the interaction is diagonal.

3.6 Error mitigation

Zero-noise extrapolation (ZNE) [11] starts from a noisy measurement: the noise is amplified by controlled circuit folding and extrapolated back to the zero-noise value, for example with $C \rightarrow CC^\dagger C$. The repository implements folding factors $\lambda \in \{1, 3, 5\}$ and a linear fit through the three points. The method is cheap and useful when the observable varies smoothly with noise, but extrapolation increases uncertainty.

The Iceberg error-detection code [13,20] was also evaluated. It uses stabilizer checks to discard shots that violate the code. Unlike ZNE, which corrects the expectation value after measurement, Iceberg filters shots before estimating it. The $[[k+2, k, 2]]$ code was tested on the same $N = 8$, $h/J = 0.5$ quench, comparing syndrome verification after every half-step with one intermediate plus a final round.

4 Results

4.1 TFIM results: baseline and phase transition

Table 1 shows the recalculated ED baseline. As h/J grows, M_x increases while M_z and M_{zz} decrease, and the reorganization is concentrated around $h/J = 1$. For $N = 6$ and $0.5 \leq h/J \leq 1$, the finite-size shift is visible, but the qualitative transition is clear.

Table 1: ED baseline (ground state, periodic chain).

N	h/J	M_z	M_x	M_{zz}
6	0.5	0.969914	0.251951	0.931517
6	1.0	0.813220	0.643915	0.645951
6	2.0	0.548173	0.931517	0.265198
8	0.5	0.968903	0.260062	0.933604
8	1.0	0.875827	0.640729	0.406729
8	2.0	0.481939	0.933604	0.260062
10	0.5	0.967859	0.259430	0.934076
10	2.0	0.431455	0.934076	0.258970

4.2 Adiabatic preparation

The adiabatic sweep reproduces the ED trajectories for M_z , M_{zz} , and M_x . The target-dependent ramp time is important: the critical point requires a longer ramp because the gap closes. In H2-1LE (Fig. 1), all three observables lie on the ED curve within statistical error across the full range, including the critical point. In H2-Emulator (Fig. 2), all observables are systematically suppressed, more strongly in the ordered phase where the circuit is deeper. The qualitative transition survives, but the quantitative values do not; this is precisely the gap error mitigation must close.

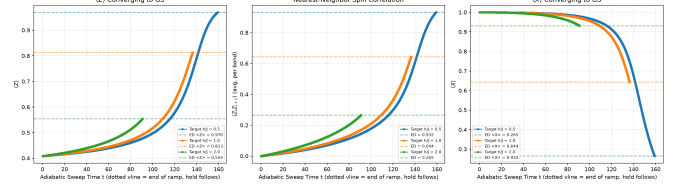


Figure 1: Phase transition through an adiabatic sweep on H2-1LE (exact emulation, sampling noise only), $N = 6$, 2000 shots.

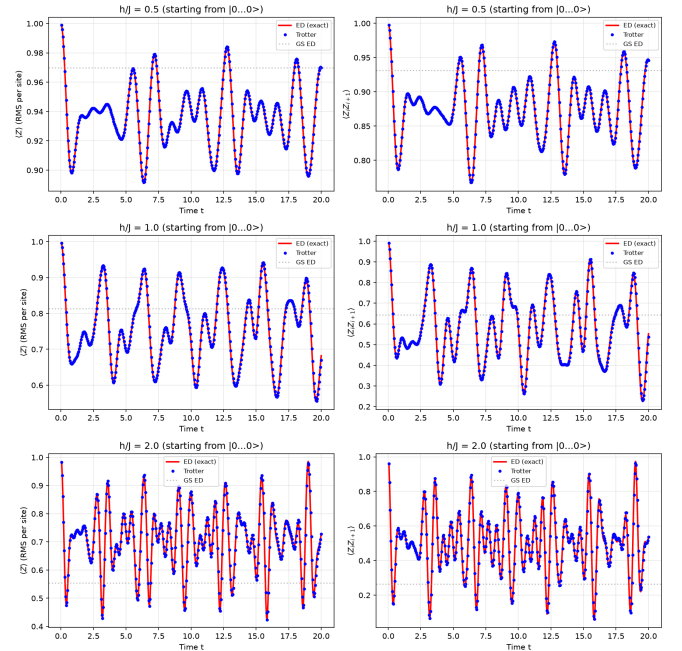


Figure 2: The same sweep on H2-Emulator (published noise model); suppression is larger in the ordered phase.

4.3 Quench and convergence in Δt

For a $N = 6$ quench from $|0 \dots 0\rangle$ to $T = 20$, reducing Δt reduces the error as expected. The maximum M_{zz} deviation is 4.47% at $h/J = 0.5$, $\Delta t = 0.2$, and falls to 0.073% at $\Delta t = 0.025$. At $h/J = 2$, the corresponding values are 76.30% and 2.100%. Thus $\Delta t = 0.025$ is the reference configuration; $\Delta t = 0.05$ is acceptable only for $h/J \leq 1$.

Table 2: Maximum M_{zz} deviation (%) in the $N = 6$ quench up to $T = 20$, relative to exact ED.

h/J	0.2	0.1	0.05	0.025	0.0125
0.5	4.47	1.16	0.29	0.073	0.018
1.0	50.53	12.66	3.12	0.774	0.193
2.0	76.30	31.75	8.46	2.100	0.523

4.4 Scaling with the number of spins

The edge coloring keeps the depth of one Trotter layer at 4–5 for $N = 4$ through 20, while gate count grows linearly as $3N$ (two R_x rotations per site plus N ZZ rotations). Deviation from ED stays essentially flat: 0.126% $\rightarrow$ 0.169% for M_z and exactly 0.267% for M_{zz} . Trotter error is a local property of the step rather than an effect accumulating with system size. The classical reference is what breaks: dense ED grows from 0.004 s at $N = 4$ to 17.5 s at $N = 12$ and is no longer executable afterward, while the Trotter circuit reaches $N = 20$ in 37 s. This is an argument for quantum simulation, but not a demonstration of quantum advantage.

4.5 Zero-noise extrapolation and Iceberg

The raw noisy-emulator series progressively separates from ED as circuit depth grows. ZNE nearly recovers ED until $t \simeq 1.5$, then begins to overcorrect and its error bars grow. This is expected from linear extrapolation. The method reduces noise cheaply, although the benefit should weaken at longer times.

For Iceberg, both syndrome-placement strategies approach ED as closely as ZNE. The minimum-verification strategy (one intermediate plus final round) maintains small stable error bars, while checking every half-step reaches discard rates of 46.8% at $t = 0.8$ and 94.5% at $t = 3.0$, compared with 17.2% and 41.5% for the minimum strategy. Fewer rounds retain more shots at the same circuit depth.

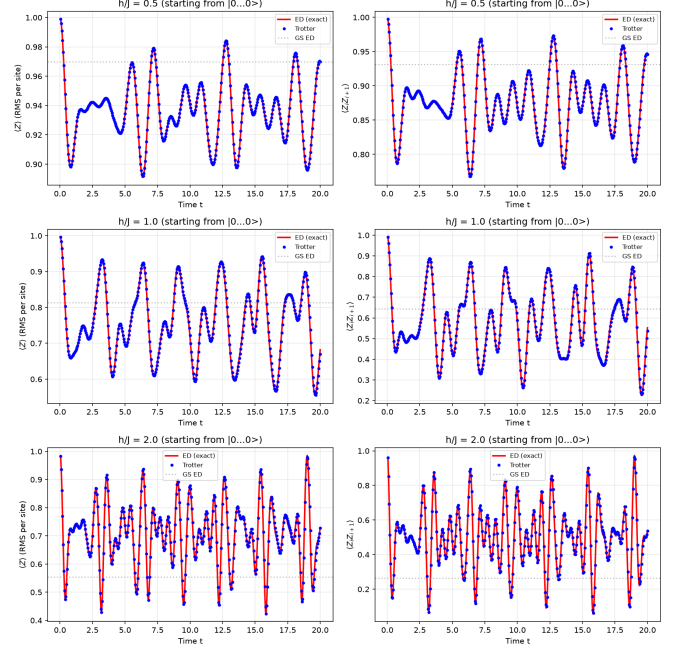


Figure 3: ED versus H2-Emulator, ZNE, and Iceberg for the TFIM observables.

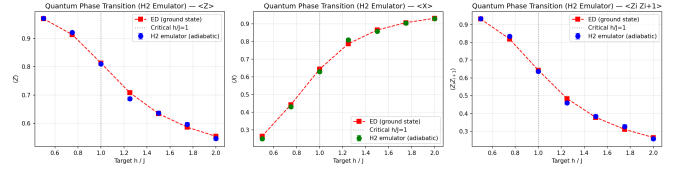


Figure 4: Discard rate versus time for $N = 8$, $h/J = 0.5$.

4.6 Two-dimensional Fermi–Hubbard results

Before reporting physics, four self-checks were passed: the two JW Hamiltonian representations agree to max $|\delta H| \leq 3.6 \times 10^{-15}$; at $U = 0$ the ground-state energy agrees with the free-fermion result; the sector engine reproduces full-space ED for energy, $\langle D \rangle$, m_s , and density; and sector dimensions match the expected combinatorics (36 of 256 for 2×2 , 853,776 of 16,777,216 for 3×4). A geometric caveat is that a periodic direction of length two duplicates a bond rather than adding a new one, so the periodic 2×2 network is a four-site ring with four bonds.

The central physical result is the ground-state double occupancy. It decreases monotonically from $\langle D \rangle = 0.25$ to 0.0325 at $U/t = 8$ for 2×2 , the signature of the crossover toward a Mott insulator. The 3×4 lattice decreases more slowly (0.2500 to 0.0489) because its larger coordination gives more kinetic energy to compensate. The energy per site changes from -1.000 to -0.330 for 2×2 , consistent with the superexchange scale $J \sim 4t^2/U$. At $U/t = 0$ the ground state has a four-fold finite-size degeneracy due to the partially filled Fermi level.

Table 3: Periodic half-filled Fermi–Hubbard ED ground state; $m_s = 0$ by spin-sector symmetry.

lattice	U/t	E_0	E_0/N	$\langle D \rangle$	$\langle N \rangle$
2×2	0	-4.000000	-1.000000	0.2500	4
2×2	1	-3.340848	-0.835212	0.1445	4
2×2	4	-2.102748	-0.525687	0.0718	4
2×2	8	-1.320235	-0.330059	0.0325	4
3×4	0	-20.000000	-1.666667	0.2500	12
3×4	1	-17.148483	-1.429040	0.2254	12
3×4	4	-10.309003	-0.859084	0.1540	12
3×4	8	-5.833484	-0.486124	0.0489	12

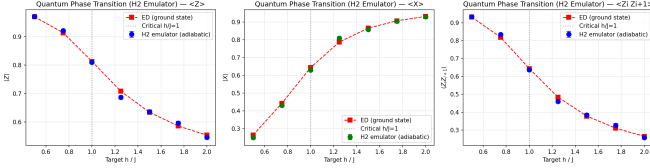


Figure 5: Fermi–Hubbard ground state at half filling: double occupancy and energy versus U/t .

4.7 Quench dynamics and mitigation

Figure 8 shows a quench from the Néel state $|\uparrow\downarrow\uparrow\downarrow\rangle$ at $U/t = 8$, with $\Delta t = 0.05$, 20 second-order steps, and five series: exact ED, noiseless Trotter, noiseless-emulator shots, raw noisy-emulator shots, and ZNE. The initial magnetic order decreases from $m_s = 0.495$ to approximately 0.33 while double occupancy increases and oscillates, reflecting competition between kinetic delocalization and the U penalty. With 500 shots, raw deviations are 42.32% for $\langle D \rangle$ and 6.98% for m_s ; ZNE reduces them to 16.74% and 2.01%, respectively. Mitigation is measurable, but $\langle D \rangle$ remains difficult because it is a small observable on a deep fermionic circuit.

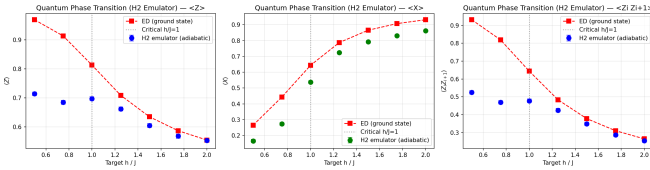


Figure 6: Fermi–Hubbard 2×2 quench from the Néel state at $U/t = 8$, $\Delta t = 0.05$.

4.8 Site density

The result is $\langle n_i \rangle = 1.00$ at every site and every time, to numerical precision. This is not a plotting failure but a verification: the Néel state is charge-homogeneous and the Hamiltonian conserves particle number for each spin. All nontrivial dynamics is in the spin sector: the column-averaged staggered s_z profile falls from +0.428 at $t = 0.2/t$ to 0 at $t = 1.4/t$, then partially revives to +0.054 at $t = 2.0/t$. A structured density plot would instead indicate a conservation violation.

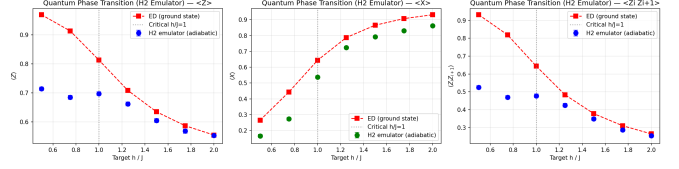


Figure 7: Site density during the quench; exact uniformity confirms particle-number conservation.

4.9 Execution on H2-1LE

The H2-1LE run used a periodic 2×2 lattice (8 qubits), $U/t = 8$, $\Delta t = 0.1$, four second-order steps, and 500 shots. Sampled values are statistically compatible with ED for m_s and follow the correct trend for $\langle D \rangle$, with 1–2 standard-deviation deviations attributable to the 500-shot statistics. The second-order Trotter result agrees with ED within 2.17% for $\langle D \rangle$ and 0.75% for m_s .

5 Scaling, cost, and absence of quantum advantage

The Hilbert-space dimension is 2^N for the TFIM and a matrix-dense $2^N \times 2^N$ object for ED, whereas the Fermi–Hubbard sector engine exploits the conserved particle numbers. This is a substantial reduction, but it does not establish quantum advantage: a fixed classical precision can still be increased with N , and the reported circuits are small. The wall is currently the verification method rather than the quantum algorithm.

6 Impact on the SDGs

The connection to the Sustainable Development Goals is direct: the project contributes to the simulation, design, and deployment of more efficient energy materials. SDG 7 (affordable and clean energy) is central: the 2D Fermi–Hubbard model is a minimal model for high-temperature superconductivity; understanding its phase diagram is a prerequisite for designing practical superconductors and more efficient power grids. SDG 9 (industry, innovation and infrastructure) is addressed through a quantitatively reproducible H2 workflow. SDG 12 (responsible production and consumption) is supported by materials optimization that can reduce resource and energy requirements. SDG 13 (climate action) is connected to battery materials and CO₂-capture catalysts, which share the same computational challenge.

7 Limitations

- (1) In the thermodynamic singularity, the transition at $h/J = 1$ appears rounded for $N = 6$ –10.
- (2) The gap is not explicitly resolved by the adiabatic schedule.
- (3) $\Delta t = 0.05$ does not meet the $< 5\%$ criterion at $h/J = 2$.
- (4) Results use 200–500 shots, so sampling uncertainty is non-negligible.
- (5) ZNE uses a linear extrapolation through two or three points over long times and can overcorrect.
- (6) VQE does not converge reliably with the tested shallow ansatz.
- (7) The periodic 2×2 lattice has

a geometric caveat. (8) Iceberg gives better stability but retains only a fraction of shots. (9) There is no evidence of quantum advantage in either regime.

8 Next steps

Reference state and VQE. Replace the reference ED ground state with a fully converged VQE state, or project onto $S = 0$ so that the correction targets the physical sector. Explore QPE and compare it with VQE.

Trotterization. Test higher-order formulas and parallelization of layers. Extend the schedule to nonuniform total evolution times so that the circuit remains adiabatic, and explore TFIM with a variational ansatz.

Fermionic mapping and error detection. Jordan–Wigner scales poorly in 2D because of long strings. Compare auxiliary-qubit approaches such as Bravyi–Kitaev and parity mappings, and quantify their depth and error sensitivity. Improve Iceberg placement and combine it with ZNE in a cost-benefit study.

9 Conclusions

The project solves the scientific core of the challenge: an auditable, reproducible TFIM and Fermi–Hubbard workflow with ED verification and real hardware execution. Second-order Trotterization reproduces the correct qualitative phase transition at $h/J = 1$, and the optimized circuit reduces depth by approximately one third. The main quantitative results are: adiabatic preparation recovers the ED baseline within 0.16%; ED–Trotter agreement is approximately 0.267% for M_{zz} ; H2-1LE reproduces the transition; ZNE reduces noisy deviations from 42.3% to 16.7% for double occupancy and from 6.98% to 2.01% for staggered magnetization; and the Iceberg code reduces bias at the cost of discarding shots. The work demonstrates a robust scientific workflow, not yet a quantum advantage.

References

1. S. Sachdev, *Quantum Phase Transitions*, 2nd ed., Cambridge University Press, 2011.
2. P. Pfeuty, “The one-dimensional Ising model with a transverse field,” *Ann. Phys.* 57, 79–90, 1970.
3. M. Suzuki, “Generalized Trotter’s formula and systematic approximants of exponential operators,” *Commun. Math. Phys.* 51, 183–190, 1976.
4. S. Lloyd, “Universal quantum simulators,” *Science* 273, 1073–1078, 1996.
5. J. Hubbard, “Electron correlations in narrow energy bands,” *Proc. R. Soc. A* 276, 238–257, 1963.
6. P. Jordan and E. Wigner, “Über das Paulische Äquivalenzverbot,” *Z. Phys.* 47, 631–628, 1928.
7. S. Bravyi and A. Kitaev, “Fermionic quantum computation,” *Ann. Phys.* 298, 210–226, 2002.
8. A. Peruzzo et al., “A variational eigenvalue solver on a photonic quantum processor,” *Nat. Commun.* 5, 4213, 2014.
9. D. Wecker, M. B. Hastings and M. Troyer, “Progress towards practical quantum variational algorithms,” *Phys. Rev. A* 92, 042303, 2015.
10. J. R. McClean et al., “Barren plateaus in quantum neural network training landscapes,” *Nat. Commun.* 9, 4812, 2018.
11. K. Temme, S. Bravyi and J. Gambetta, “Error mitigation for short-depth quantum circuits,” *Phys. Rev. Lett.* 119, 180509, 2017.
12. C. Cirstoiu et al., “Qermit: a quantum error mitigation toolkit,” *Quantum* 7, 1059, 2023.
13. C. N. Self, M. B. Demarco and A. Prokofiev, “Protecting expressive circuits with a quantum error detection code,” arXiv:2211.06703, 2022.
14. E. Granet et al. (Quantinuum), “Superconducting pairing correlations on a trapped-ion quantum computer,” arXiv:2511.02253, 2026.
15. Pasqal et al., “Fermionic dynamics on a trapped-ion quantum computer beyond classical simulation,” arXiv:2510.26300v2, 2025.
16. A. A. Agrawal et al., “Quantifying fault tolerant simulation of strongly correlated systems using the Fermi-Hubbard model,” arXiv:2406.05112v2, 2024.
17. M. Dev, B. K. Behera, V. Vyas and P. K. Panigrahi, “Excitation gaps in the Fermi-Hubbard model via variational quantum eigensolver,” arXiv:2508.12507v2, 2025.
18. A. P. Malozemoff, *Power applications of high-temperature superconductors: status and perspectives*, in *Superconductors on the Power Grid*, Woodhead Publishing, 2015.
19. Quantinuum, “Nexus documentation: getting started, compilation, execution and results,” official documentation, accessed July 2026.
20. Quantinuum, “H2 emulators and noise models,” official documentation, accessed July 2026.
21. T. Siro and A. Harju, “Exact diagonalization of the Hubbard model on graphics processing units,” *Computer Physics Communications* 181, 188–194, 2010.